Contamination models: declared failure modes for belief fusion I

Robust belief fusion only earns its keep when some agents are wrong. The active-inference community has built ensembles that coordinate by sharing beliefs and observations (Friston et al. 2024; Heins et al. 2024; Albarracin et al. 2022; Kaufmann et al. 2021), but it has assumed those beliefs are trustworthy in the cited modeled protocols: fusion is treated as exact-Bayes pooling of well-calibrated reports. The robust-Bayes and federated-learning literatures (McMahan et al. 2017; Ashman et al. 2022; Mildner et al. 2025) have, in turn, studied robustness to corrupted clients under their declared settings, but outside the generative-model-bearing POMDP setting. This section defines the corruption process that lets us test fusion robustness inside the active-inference colony — the experimental complement of the robust aggregation rule of sec. 5.8.

Corruption process for adversarial belief broadcasts I

In the sentinel world a healthy sentinel reports a well-calibrated categorical over the creature's location; a contaminated one reports something corrupted. `contamination.contaminate` manufactures the corrupted reports. Every corruption is a convex mixture of the agent's belief b with a corruption target t, governed by a single rate $r \in [0, 1]$,

$$\tilde{b} \;=\; (1 - r)\, b \;+\; r\, t, \tag{1}$$

so the experiments sweep exactly one knob. The convex form of eq. 1 gives a clean limit and is the anchor of the suite (ISC-26): at $r = 0$ every corruption kind returns the input belief unchanged, so contamination is a strict, continuous departure from the uncorrupted Friston belief-share — never a discontinuity. This section defines the three core corruption targets t, each capturing a distinct failure of a federated agent. Geometrically the three are three landmarks of the probability simplex — a wrong vertex (`confident_wrong`), the flat centroid (`uniform`), and a random interior point (`label_noise`) — so the mixture of eq. 1 drags an honest belief toward a qualitatively different destination in each case. Two further mechanisms (`byzantine` and `drift`) extend the same convex-mix contract and are introduced in the extended-methods supplement (sec. 12.2).

`confident_wrong` — the adversarial sentinel. This is the lookout that points to one wrong cell and insists on it with total certainty. The target is a one-hot spike on a wrong state, $t = \mathrm{onehot}(s_{\mathsf{wrong}})$, so $\tilde{b}$ is mixed toward a confident, mistaken delta. Callers choose s_{wrong} explicitly; the verdict sweep of sec. 5.22 fixes it once per colony as the state diametrically opposite the true state on the location grid, held constant across the entire rate sweep, rather than deriving it from the agent's current belief. At $r = 1$ this is a pure delta on the wrong cell. This is the saboteur that is *sure* and *mistaken*: exactly the agent that robust aggregation must reject.

`label_noise` — the miscalibrated sentinel. This is the lookout with a scrambled sensor: it is not lying toward any particular cell, only diluting every honest report with the same fixed sprinkle of noise. The target is a fixed noisy categorical drawn once from a $\mathrm{Dirichlet}(1)$ (a random but valid pmf), modeling a sentinel whose report is partly random rather than adversarial. Because the noisy target is drawn once and then held fixed across the rate sweep, the corruption has no direction to exploit and no single cell to veto — the robust pool meets diffuse degradation, not a targeted attack.

`uniform` — the apathetic sentinel. This is the lookout that shrugs: it has lost track of the creature and calls every cell equally likely. The target is the maximum-entropy uniform pmf $t = (1/n_s)\mathbf{1}$, modeling a saturated sentinel that has lost all information. At $r = 1$ it reports uniform, contributing no evidence to the pool rather than actively pulling it toward a wrong cell.

All three share the contract of eq. 1 and require an explicit seeded generator — `label_noise` uses it to draw the noisy target — so every contaminated report is reproducible. The grid of rates the sweep uses, $\{0, 0.225, 0.45, 0.675, 0.9\}$, deliberately stops below the pure-veto limit $r = 1$, where a fully-confident wrong delta forces *every* pooling rule's accuracy to zero and the robust-versus-naive contrast vanishes.

A contaminated report feeds the colony in distinct places, and the honesty contract of sec. 3.3 turns on keeping them separate.

At the *server* (the aggregation step of sec. 5.8) a contaminated belief enters `robust_aggregate`, the iteratively-reweighted pool that discounts each agent by $\exp(-c\,\mathrm{KL}(q_n \,\|\, q))$. A confidently-wrong agent sits far from the emerging consensus, earns a small effective weight, and is suppressed. This is the *heuristic* axis: its only proven property is that at $c = 0$ it recovers the project's naive log-linear pool exactly (eq. 7, Theorem 5). Under the qualified bridge of sec. 5.8, that pool specializes Eq. 7's message-combination term rather than the complete source protocol. The robustness-sweep figures (fig. 12, fig. 13) illustrate this heuristic's behavior —

including the per-agent influence weights that drop the saboteurs — but they do not certify a per-agent guarantee.

At the *client* (the per-agent generalized-Bayes update of sec. 5.7) contamination is what the bounded β-loss (eq. 4) and rcce-loss (eq. 5) are designed to survive: a single corrupted observation with $p(o) \to 0$ drives the unbounded NLL to dominate the posterior, whereas the bounded losses cap its influence. This is the source-theorem-backed axis: the FedGVI guarantee (Mildner et al. 2025) is inherited only under the source theorem's matching loss, divergence, and regularity assumptions. The federated logistic-regression baseline of sec. 5.22 applies this same client mechanism to flipped-label contamination (fig. 16); it is the conjugate Bernoulli analogue of the categorical client update, and its robustness is the per-client loss, not the server reweighting.

How contamination meets the three robustness axes I

No figure, statistic, or sentence in this manuscript grants the server-side heuristic the per-client bounded-influence guarantee; contamination is the common stressor against which the three axes are kept distinct.